

cert-manager - Past, Present and Future

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BUILDING FOR THE ROAD AHEAD

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The Past: Project History



May 2016: kube-lego open-sourced by Jetstack

October 2017: v0.1.0 release of cert-manager

September 2020: cert-manager v1.0.0 released

November 2020: accepted as CNCF Sandbox project



The Present: Incubation

October 2022: cert-manager accepted
as a CNCF incubating project!



The Present: Incubation



CLOUD NATIVE
COMPUTING FOUNDATION

INCUBATING PROJECT

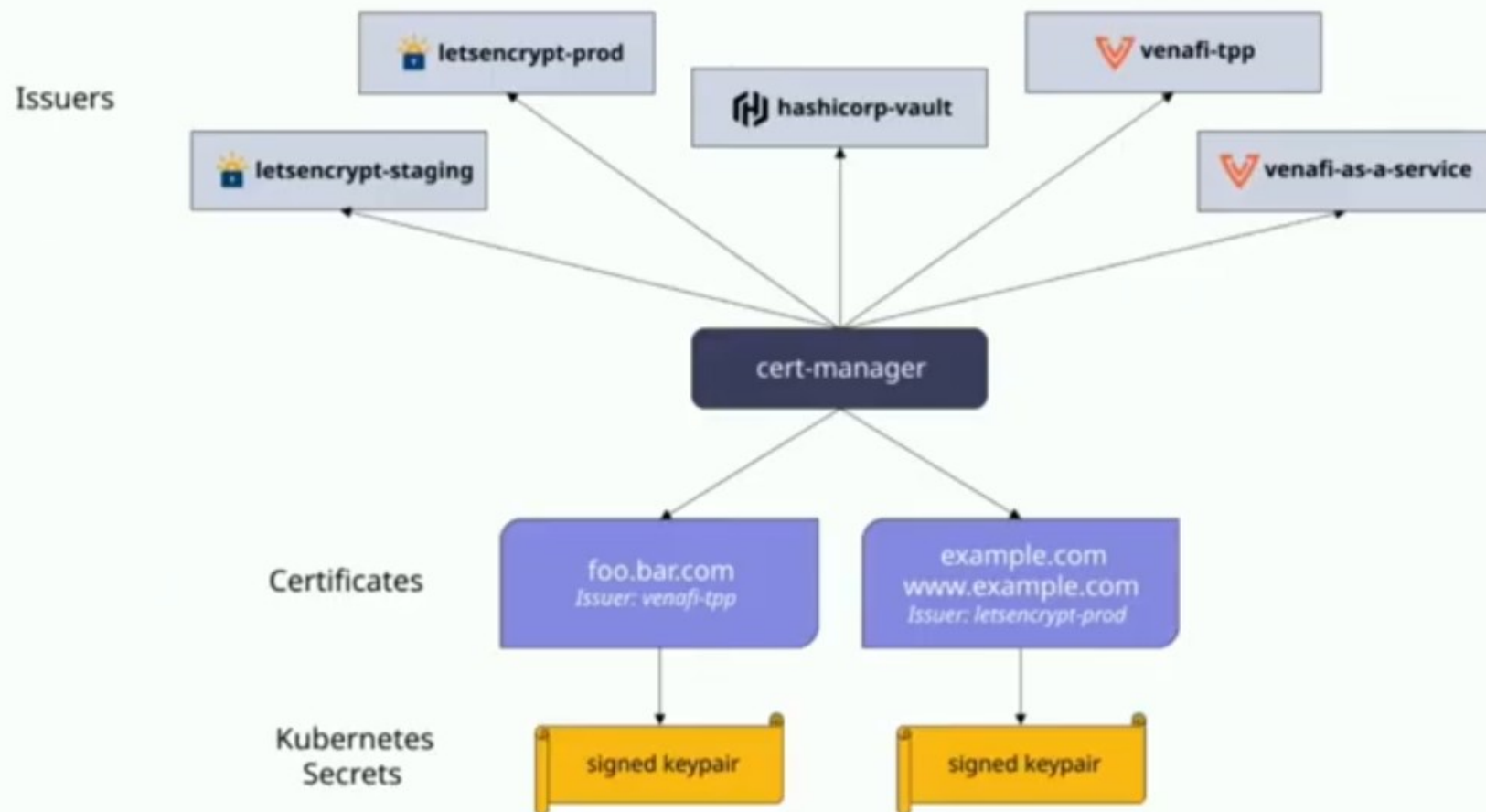
The Present: What's cert-manager?



- Cloud-native management of X.509 certificates
- Simple workflow
- Automatic lifecycle management
- Extensible issuers and policies
- 9.5k+ GitHub Stars!



The Present: High-level Overview



The Present: Certificate Issuance



- Automated ACME challenge solving for publicly trusted certificates
- Supports ACME (Let's Encrypt, Smallstep step-ca), Hashicorp Vault, Venafi
- Automated renewal of short-lived X.509 certificates
- Extensible - easy to support new issuers and DNS providers



The Present: Security



- cert-manager is critical security infrastructure!
- We're early adopters of Project sigstore; all our main release artifacts are signed with cosign
- "Self-certified" SLSA 2 compliance, almost level 3



sigstore
cosign



The Present: Policy



- cert-manager isn't just a single app, it's a whole project!
- approver-policy: who requested that certificate? Should it be allowed?
- Short-lived certificates for mTLS credentials
- Private Key rotation policies



The Present: Integrations



- Custom approval policy integration, e.g. OPA
- Integrates with other cloud native projects:
 - Istio (replacing citadel)
 - Linkerd
 - sig-network Gateway API (Traefik, Contour, Istio)
 - Prometheus (rich metrics for certificate state)
- CSI driver integration
 - SPIFFE (X.509)



The Future: Integrations



- An integration with your project?
- We'd love to talk to you!
- Drop by our booth in the pavilion!



The Future: Roadmap



- Increase collaboration with other CNCF projects
 - Service Meshes
 - SPIFFE/SPIRE
- Continue to improve developer experience
- Better support for X.509 features - extended attributes, CRLs/OCSP
- Explore other key management opportunities e.g. code signing



The Future: Roadmap File



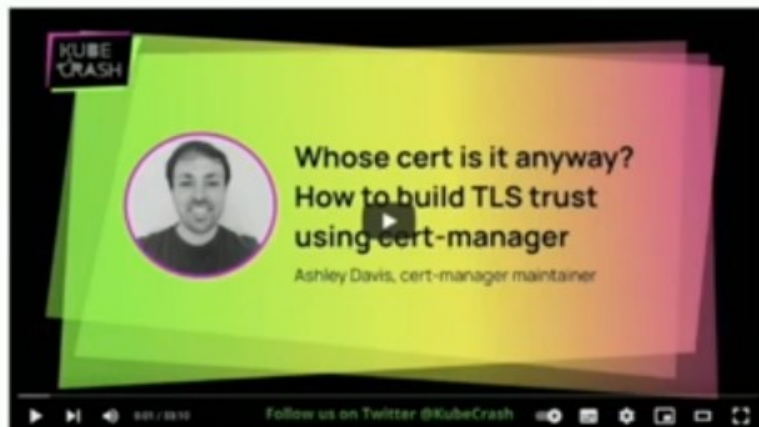
- Roadmaps are important for open source projects
- Our public roadmap is available in the cert-manager repo
- No promises on timelines, but we do our best!



The Future: trust-manager



- Getting a certificate is only half the TLS problem!
- We need to know how clients can verify certificates
- trust-manager seeks to solve this for TLS
- NB: There's upstream work on this: [trust anchor sets](#)



The Future: You?



- We get a lot of issues and PRs (that's great!)
- There's a tonne to do!
- We'd love more maintainers to get stuck in with us
- If you're interested, please talk to us
- Open source maintainership experience not required!
- We can help



Visit Our Booth!

- Drop by the cert-manager booth if you haven't already!
- Get your own souvenir certificate
- We'd love to have a TLS handshake (or elbow bump) with you!





- <https://cert-manager.io> - cert-manager website
- <https://github.com/cert-manager> - GitHub org
- [Meetings](#) - Details on when we have public meetings
- [Slack](#) - Chat with us on Kubernetes slack
- cert-manager-maintainers@googlegroups.com - email the maintainer team!





Thank you!